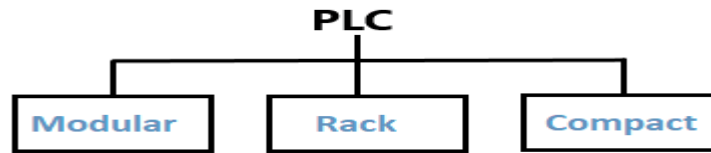


TOPIC 1: INTRODUCTION TO PLC'S (Cont'd)

Types of PLC Designs (constructions/units)

There are three types of PLC, which are shown below



a) Unitary/Compact PLC Design

A Unitary PLC contains all the features of a basic system in one compact unit. They are supplied as an integral compact package complete with power supply, processor, memory, and input/output units. They are fitted directly to the component or machine that they are controlling.



Advantages

- ✓ They are small and compact.
- ✓ They hold all the basic components in one unit.
- ✓ They allow portable and easy access.
- ✓ They are usually the cheapest type of PLC.

Disadvantages

- ✓ They cannot be expanded.
- ✓ If any feature fails then the whole unit has to be replaced.
- ✓ They are very simple and basic.

Applications

Unitary PLC's can be used for any application that does not require a lot of inputs or outputs. For example a car park barrier, this would not use many inputs/ outputs and would only require a simple program.

b) Modular PLC Design

Modular PLC's are a system of modules that can be slotted together to build up a system. The modular type consists of separate modules for power supply, processor, etc., which are often mounted on rails within a metal cabinet. Other types of module can be attached as well as extra input and output modules to increase the capacity or to cope with changes in hardware system

Advantages

- ✓ The amount of input and output terminals can be expanded to cope with any changes to the hardware system.
- ✓ If any feature fails then only that part has to be changed saving on cost.

Disadvantage

- ✓ They are expensive compared to unitary PLC's.

Applications

Modular PLC's are useful for applications where there would be a lot of inputs and outputs because more input/output modules can be added if needed. This type of PLC is widely used in the manufacturing industry to control process lines.

c) Rack-Mounted PLC Design

The design and characteristics of a rack mounted system are very similar to a modular system however these modules are on standard cards which then slot together into a rack inside a control cabinet. These modules communicate via the rack. The slots in the rack PLC communicate with each other connected by the standard network or bus.

Rack PLC



Advantages

- ✓ They are easily modified or expanded.
- ✓ They have more I/O points than any other type of PLC.
- ✓ If any feature fails then only that part has to be changed saving on cost.

Disadvantages

- ✓ They are usually the most expensive of the three PLC's

Applications

Like the modular PLC this type of system is widely used in the manufacturing industry. This is mainly because of the amount of input/output points that the system can contain but also because it has the ability to communicate with other networks.

PLC Manufacturers

The industries that make PLC are listed below:

- *Siemens*: It is the German multinational company that manufactures PLC. It is well known and most used PLC all over the world.
- *Rockwell Automation*: It is an American company that also took over the Allen Bradley brand. It is the largest manufacturer of PLCs and well known in the United States.
- *ABB (Asea Brown Boveri)*: The ABB is a Swiss multinational company that provides PLCs within vast industries and applications.
- *Omron*: It's a Japan-based electronics company.
- *General Electric*: It's an American based PLC company.
- *Delta*: It deals with all types of automation machines. It is a Taiwan- based PLC manufacturer.
- *Schneider*: The Schneider is a French-based PLCs manufacturer.
- *Mitsubishi*: It is a Japanese MNC group that manufactures PLC.
- *Koyo Electronics Corporation Limited*: It manufactures the DirectLogic PLC brand. It is a Japanese based industry.

TOPIC 5: CALIBRATION SOFTWARE

Calibration is the process in which an instrument's accuracy is compared with a known and proven standard. Calibration software refers to applications that automate all or part of a calibration process via computer control. Calibration software also allows the user to manage their calibration and asset data.

Importance of calibration software

a) Consistency

Software automation ensures that the calibration can be performed exactly the same way by multiple operators in multiple locations, and over time. This improves the quality of the results, reduces errors and standardizes methods.

b) Analyze results and make process improvements

Using calibration software saves data entry time, minimizes the risk of errors and securely stores the data. The software gives the ability to analyze calibration results and determine optimal calibration intervals. Digital records provide a path for post-calibration data analytics that would be practically impossible with paper records.

c) Efficiency

Automating the calibration process allows operators to set up tests and then go on to perform other tasks, making more efficient use of their time. Calibrations are typically completed much more quickly, saving time and money. If the software is capable of calibrating multiple units under test simultaneously, automation increases throughput as well.

d) Documentation and reports

Calibration automation software typically includes features for documenting calibration procedures, storing calibration data, and producing reports, allowing the operator to eliminate paper records or spreadsheets. Because calibration software does such a wonderful job of keeping accurate records of all parts of the calibration process, it can also help you comply with a wide variety of quality standards.

e) Be compliant

Industry regulations, internal standards of excellence and insurance requirements may require documented proof of calibration. Calibration certificates can automatically be generated from the calibration results and formatted to display relevant information needed to be compliant.

Factors to consider when selecting calibration software

- a) *Ability to handle the required workload:* Make sure the software selected is compatible with the calibration instruments and the intended workload. Choose a solution that is scalable and that can grow as the needs change.
- b) *Features:* Select calibration software that has all the features needed for the calibration under consideration.
- c) *Ease of use:* Check whether the software is designed by someone who understands the calibration process and is simple to use
- d) *Ability to help run the lab more efficiently:* Good software should help reduce costs and become more productive in the lab. It should help do more calibrations in the same or less amount of time and enable fast and repeatable calibration
- e) *Flexibility:* Should be possible to install on single workstations as well as company-wide networks and be accessible by multiple users from different locations
- f) *Regulatory compliance:* The software should be able to support the traceability and record-keeping requirements of modern quality and accreditation standards without having to write new programs, set up new databases or test and documents the system.
- g) *Ability to support unified data management across different metrology disciplines:* Since one may choose different applications specialized in electrical, temperature, pressure, flow, or dimensional calibration, the software should allow for management of all these data in one database
- h) *Availability of training:* A good calibration software is one that its developer provides training to help the owner and the staff get up and running quickly
- i) *Kinds of support are available:* The manufacturer should provide contacts in case of questions or problems in using the software. Some software developers offer premium support programs that help maximize productivity and minimize downtime. Look for other types of available support as well, including user groups, online forums, technical and application notes and other informative materials.
- j) *Manufacturer:* Select calibration software produced by a reputable company that will still be in the calibration business by the time you need support. Pick a developer who is committed to the long-term future of its software applications and will provide regular updates and upgrades.

Types of Calibration

In general terms, there are two types of Calibration systems namely;

- ❖ Hand held calibration system
- ❖ Computer based calibration

1. Hand-Held Device

This is a mobile/portable device which is a small, handheld computing device, typically having a display screen with touch input and/or a miniature keyboard and weighing less than 0.91kg. A handheld computing device has an operating system (OS), and can run various types of application software. Most

handheld devices can also be equipped with Wi-Fi, Bluetooth, and GPS capabilities that can allow connections to the Internet and other Bluetooth-capable devices, such as an automobile or a microphone headset. It delivers messages and performance data to the operator and is used to support installation, configuration, provisioning, calibration and maintenance and network performance.

Advantages of handheld calibration

a) No process interruption

The main advantage of handheld calibration over other traditional methods is that it allows meter verifications to be carried out directly in the process without additional costs for removal of the instrument or process interruptions. As a result, downtime is minimized and critical processes can be verified and optimized efficiently. By supporting and facilitating regular on-site verification, handheld equipment helps users to quickly diagnose any failures and to swiftly remedy the situation.

b) Time and cost savings

Device verification using handheld equipment requires a maximum of 15-30 minutes per instrument. The device does not need to be sent away to the calibration center and production can, therefore, resume faster than with any other method. After the process has been completed, the direct uploading of the device parameters avoids time-consuming configuration. This method helps achieve optimum availability of plant equipment. The production does not need to be suspended resulting in considerable savings.

c) Complete on-site verification

Handheld electronic verification not only checks the accuracy of the device under test, but also performs a complete check of the entire measurement chain.

d) Simulation of the process

As safety during operation is considered a 'must' for plant operators, testing the safety and functionality of equipment in the process is often indispensable. Simulation of the measuring signals during calibration or verification processes can achieve that. Handheld calibrators can simulate process states, in flow applications for example, even without real flow. Handheld verification devices can simulate different flow rates in the process; high and low limit values, receiving signal chains (operation of valves or control loops) and different flow behavior in piping, for example, such as flow curves in bottling machines.

2. Computer-Based Calibration System

A computer-based calibration system uses specialized calibration software that automates the calibration process by interacting with the instruments through a computer. It allows for more complex, precise, and efficient calibration processes when compared to handheld devices. It's ideal for environments that require high-volume, highly accurate, and traceable calibration processes.

Advantages:

- i. *Automation and Accuracy:* Reduces human error by automating calibration steps, ensuring consistent and precise results.
- ii. *Efficient Data Management:* Stores large amounts of calibration data digitally, making analysis, access, and reporting easier.

- iii. *Scalability*: Can handle multiple instruments at once, making it ideal for high-volume calibration environments.
- iv. *Customization and Flexibility*: Allows for tailored calibration procedures and can grow with changing needs.
- v. *Traceability and Compliance*: Meets regulatory standards with automatic record-keeping and traceable calibration data.
- vi. *Cost Efficiency*: Reduces manual labor and optimizes calibration intervals, saving time and money.
- vii. *Integration*: Works with other systems like asset management or lab management tools for unified data handling.

Disadvantages:

- i. *Initial Cost*: The setup cost for a computer-based system can be high, including software licenses, hardware, and training.
- ii. *Technical Expertise*: The use of complex software and equipment requires skilled personnel who are trained to operate and maintain the system.
- iii. *Potential for Downtime*: If the system experiences a technical failure, it could cause downtime, particularly if the support and maintenance are not readily available.

Calibration Process

A calibration professional performs calibration by using a calibrated reference standard of known uncertainty to compare with a device under test. The operator records the readings from the device under test and compares them to the readings from the reference source. He or she may then make adjustments to correct the device under test.

Instrument calibration has to be repeated at prescribed intervals because the characteristics of any instrument change over a period of time. Factors deciding the frequency of calibration:

- ✓ usage rate
- ✓ conditions of use
- ✓ skill level of personnel
- ✓ degree of accuracy expected
- ✓ costs of calibration

Examples of calibration softwares

1. *ProCalV5*: A popular software used for managing calibration processes, asset tracking, and compliance reporting. It supports multi-device calibration and generates automatic certificates.
2. *GAGEpack*: Designed for managing measurement devices and calibration records. It offers scheduling, history tracking, and detailed reporting for regulatory compliance.
3. *CompuCal*: Known for its user-friendly interface, CompuCal provides full calibration management with automated scheduling, asset management, and comprehensive reporting tools.
4. *MET/CAL*: A powerful software suite for automating calibration procedures. It's widely used in industries that require high accuracy, like aerospace and defense, and integrates with Fluke instruments.

5. *Calibrate*: A web-based software designed for managing calibration workflows and asset tracking. It offers flexibility for small businesses as well as large organizations, with cloud and on-premises deployment.
6. *EasyCal*: Developed by Time Electronics, EasyCal automates calibration routines for various instruments, generating compliance-ready reports and certificates while optimizing workflow efficiency.